Movie Plot and Character Intelligence using Retrieval-Augmented Generation

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Github: https://github.com/Deepak4557/Movie-Intelligence-Character-Rag

Abstract

This project presents a Retrieval Augmented Generation (RAG) system designed to answer complex questions about movies by understanding plot structure, character motivations, and emotional depth. Traditional language models often generate inaccurate or shallow responses due to lack of grounding in real data. To address this, we built a RAG pipeline that retrieves relevant information from a curated dataset of five movies, including plot summaries, character analyses, themes, and dialogues. The retrieved context is then used by three open-source language models Llama3, Mistral, and Phi3 to generate accurate and meaningful answers. We evaluated the models based on accuracy, reasoning ability, emotional understanding, and hallucination. Experimental results show that the RAG approach significantly improves response quality and reduces hallucinations. Among the models tested, Phi3 achieved the best overall performance, demonstrating that even smaller models can produce strong results when combined with retrieval-based grounding.

1. Introduction

Understanding movies requires more than simple factual knowledge it involves interpreting complex narratives, character motivations, emotional depth, and thematic elements. Modern large language models (LLMs) have shown impressive capabilities in natural language understanding; however, they often struggle with domain-specific reasoning tasks such as analyzing fictional narratives. In particular, when asked questions about movies, these models tend to produce hallucinated or generic responses due to the lack of grounding in reliable and relevant data sources (Ji et al., 2023)."

One of the key limitations of traditional LLMs is their reliance on pre-trained knowledge, which may be incomplete or outdated. As a result, they are unable to consistently provide accurate explanations of character behavior, emotional dynamics, or underlying themes in films. This limitation becomes more significant when dealing with reasoning-based questions such as *"Why did the Joker become a villain?"* or *"How does guilt affect Cobb in Inception?"*, which require contextual understanding rather than surface-level information.

To address these challenges, this project proposes a Retrieval-Augmented Generation (RAG) framework for movie analysis. RAG enhances the capabilities of LLMs by combining information retrieval with text generation. Instead of relying solely on

internal knowledge, the system retrieves relevant information from a curated dataset and uses it as context for generating responses. This approach ensures that the generated answers are grounded in actual data, leading to improved accuracy and reduced hallucination (Lewis et al., 2020).

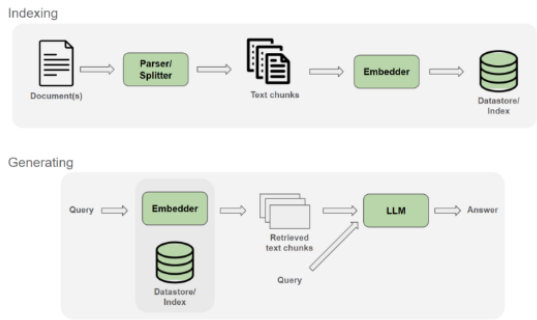

Figure 1: Overview of the RAG-based movie question answering system

In this work, we construct a domain-specific knowledge base consisting of five well-known films *The Dark Knight*, *Joker*, *Interstellar*, *Iron Man*, and *Inception*. The dataset includes plot summaries, character analyses, themes, and dialogues, enabling a rich understanding of each movie. The system processes this data using text chunking and embedding techniques, storing it in a vector database for efficient retrieval.

We implement and evaluate our RAG system using three open-source language models: Llama3, Mistral, and Phi3. The models are compared across multiple evaluation metrics, including accuracy, reasoning ability, emotional understanding, and hallucination. Through this comparative study, we aim to understand how different models perform when augmented with retrieved context.

The contributions of this project are as follows:

- Development of a RAG-based system for deep movie understanding
- Creation of a multi-dimensional movie dataset
- Evaluation of multiple LLMs on reasoning-intensive questions
- Analysis of how retrieval improves response quality and reduces hallucination

Overall, this work demonstrates that integrating retrieval mechanisms with language models significantly enhances their ability to handle complex, domain-specific tasks such as movie analysis.

2. Background

Recent advancements in Natural Language Processing (NLP) have been largely driven by the development of Large Language Models (LLMs). These models, such as Llama, Mistral, and Phi, are trained on massive amounts of text data and are capable of generating coherent and contextually relevant responses. LLMs have demonstrated strong performance in tasks such as text generation, summarization, and question answering. However, despite their capabilities, they have inherent limitations. One of the most significant challenges is *hallucination*, where the model generates information that is incorrect, fabricated, or not grounded in factual data. Additionally, LLMs rely on pre-trained knowledge, which may be incomplete, outdated, or not specific to a particular domain, such as movie narratives.

To address these limitations, Retrieval-Augmented Generation (RAG) has emerged as an effective approach that combines information retrieval with text generation. In a RAG framework, instead of

relying solely on the model's internal knowledge, relevant information is first retrieved from an external knowledge source and then provided as context to the model during response generation. This allows the model to produce answers that are grounded in actual data, improving both accuracy and reliability. RAG is particularly useful for domain-specific tasks, where precise and context-aware responses are required.

A key component of RAG systems is the use of *text embeddings*. Embeddings are numerical vector representations of text that capture semantic meaning. By converting textual data into embeddings, the system can measure similarity between different pieces of text based on meaning rather than exact keyword matching. This enables more effective retrieval of relevant information when a query is provided.

To efficiently store and search these embeddings, vector databases such as FAISS (Facebook AI Similarity Search) are used. FAISS enables fast and scalable similarity search over large collections of vectors, making it suitable for real-time retrieval in RAG systems. When a user query is received, it is converted into an embedding and compared against stored vectors to retrieve the most relevant chunks of information.

RAG vs Without RAG

Traditional LLM-based systems that do not use retrieval rely entirely on their pre-trained knowledge. While this allows for fluent and general responses, it often leads to issues such as hallucination, lack of domain-specific accuracy, and shallow reasonning. For example, when asked complex questions about movies, a standalone LLM may generate generic or incorrect explanations because it does not have access to specific plot details or character insights at runtime.

In contrast, RAG-based systems enhance LLM performance by incorporating external knowledge during inference. Instead of guessing, the model retrieves relevant information from a curated dataset and uses it to generate responses. This results in more accurate, context-aware, and meaningful answers. RAG also signif- icantly reduces hallucination, as the responses are grounded in real data. Furth- ermore, it enables smaller models to perform competitively with larger models by supplementing them with relevant contextual information.

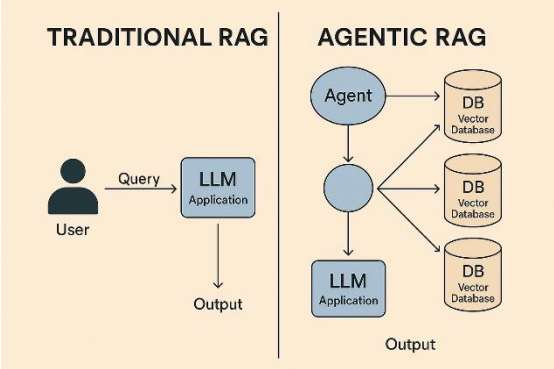

Figure 2: Rag vs Without Rag Architecture comparison

Overall, the integration of retrieval mechanisms with language models prov- ides a powerful framework for improving performance in knowledge-intensive and reasoning-based tasks, such as movie analysis.

3. Methodology

3.1 Overview

In this work, we develop a Retrieval-Augmented Generation (RAG) system for deep movie understanding. The system is designed to answer complex questions related to movie plots, character motivations, and themes by combining information retrieval with language model generation (Lewis et al., 2020).

Instead of relying only on the internal knowledge of language models, our approach retrieves relevant information from a curated movie dataset and uses it as context to generate accurate and meaningful responses. This helps reduce hallucination and improves reasoning quality.

3.2 System Architecture

The overall architecture of the system follows a standard RAG pipeline, consisting of data processing, embedding generation, retrieval, and response generation.

The pipeline can be summarized as:

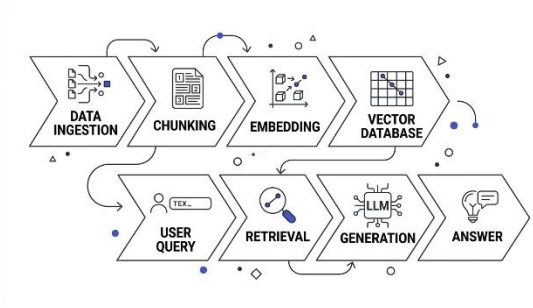

Figure 3: System Architecture

First, raw movie data is collected, including plot summaries, character descriptions, themes, and dialogues. This data is then preprocessed and divided into smaller text chunks to make it easier for the system to process and retrieve relevant information.

Each text chunk is converted into a numerical vector representation (embedding) using a pre-trained sentence transformer model. These embeddings capture the semantic meaning of the text.

The embeddings are stored in a FAISS vector database, which allows efficient similarity search. When a user submits a query, the system converts the query into an embedding and retrieves the most relevant text chunks based on semantic similarity.

These retrieved chunks are then combined to form a context, which is passed to the language model. The language model uses this context to generate an answer that is grounded in actual movie data rather than relying on prior knowledge alone.

3.3 Data Processing and Chunking

To ensure efficient retrieval, the collected movie data is split into smaller overlapping chunks. This is done using a recursive character-based text splitter with a fixed chunk size and overlap.

Chunking helps preserve context while ensuring that each segment remains manageable for embedding and retrieval. The overlap between chunks ensures that important information is not lost at boundaries.

3.4 Embedding and Vector Storage

We use a pre-trained sentence transformer model (all-MiniLM-L6-v2) to generate embeddings for each text chunk. These embeddings encode the semantic meaning of the text in a numerical format (Reimers and Gurevych, 2019).

The embeddings are stored in a FAISS vector database, which enables fast and efficient similarity search. This allows the system to retrieve the most relevant information in real time when a query is provided (Johnson et al., 2017).

3.5 Retrieval and Generation

When a user query is given, the system performs semantic retrieval to identify the most relevant chunks from the vector database. These chunks are then combined into a context.

The context, along with the user query, is passed to the language model. The model generates a response based on both the query and the retrieved information. This ensures that the generated answers are accurate, context-aware, and grounded in real data.

3.6 Language Models

We evaluate three open-source language models in our system:

- LLaMA 3
- Mistral
- Phi-3

These models are deployed locally using Ollama. Each model receives the same retrieved context and query, allowing us to compare their performance under identical conditions.

3.7 Summary

Overall, our methodology combines structured data retrieval with powerful language models to create a system capable of deep reasoning over movie content. The use of RAG ensures improved factual accuracy, reduced hallucination, and better understanding of character motivations and themes.

4. Dataset

4.1 Overview

In this project, we created a custom dataset focused on movie understanding. The dataset was designed to support deep reasoning tasks such as analyzing character motivations, emotional behavior, and thematic elements of movies.

Instead of using a standard benchmark dataset, we manually curated content to ensure that the system has rich and meaningful information for retrieval.

4.2 Movie Selection

We selected five well-known and Narratively rich movies:

- *The Dark Knight*
- *Joker*
- *Interstellar*
- *Iron Man*
- *Inception*

These movies were chosen because they contain complex characters, strong emotional depth, and meaningful storylines. This makes them suitable for evaluating deep reasoning and psychological understanding.

4.3 Data Sources

For each movie, we collected data from multiple sources to ensure diversity and completeness:

- Movie plot summaries (e.g., Wikipedia, IMDb)
- Movie scripts (where available)
- Character descriptions and analysis
- Dialogue-based content

This multi-source approach helps the system understand both factual events and deeper narrative elements.

4.4 Data Components

Each movie in the dataset includes the following components:

- **Plot Summary**: Provides the overall storyline
- **Character Analysis**: Explains motivations, personality, and behavior
- **Themes**: Highlights the central ideas of the movie
- **Dialogues**: Captures important interactions and emotional expressions

By combining these elements, the dataset provides a comprehensive understanding of each movie.

4.5 Data Preparation

All collected data was combined into a single text corpus. The text was then pre processed and structured for use in the RAG pipeline.

To enable efficient retrieval, the dataset was divided into smaller overlapping chunks. These chunks were later converted into embeddings and stored in a vector database for semantic search.

4.6 Dataset Characteristics

The dataset is relatively small in size but rich in semantic information. It is specifically designed for:

- Character-level reasoning
- Emotional understanding
- Theme analysis
- Context-aware question answering

This makes it well-suited for evaluating the effectiveness of Retrieval-Augmented Generation systems in narrative domains.

5. Question Design

5.1 Overview

To evaluate the effectiveness of our RAG system, we designed a set of domain-specific questions focused on deep movie understanding. These questions go beyond simple factual queries and instead require reasoning, emotional interpretation, and contextual analysis.

The goal of these questions is to test how well the system can interpret character motivations, psychological behavior, and thematic elements using retrieved information.

5.2 Design Strategy

The questions were carefully designed based on the following principles:

- **Reasoning-focused**: Questions require explanation of *why* events happen
- **Emotion-aware**: Focus on character feelings and psychological depth
- **Context-dependent**: Answers require understanding of story context

- **Non-factual**: Avoid simple questions like “who” or “when”

This approach ensures that the evaluation reflects real-world usage of language models in narrative understanding tasks.

5.3 Question Set

We developed the following set of questions:

1. *Why did Joker become a villain?*
2. *What motivates Tony Stark to become a hero?*
3. *What is the central theme of Interstellar?*
4. *How does guilt affect Cobb in Inception?*
5. *Why did Harvey Dent become Two-Face?*
6. *How does society influence Arthur Fleck’s transformation into Joker?*
7. *Why does Batman take the blame for Harvey Dent’s actions?*
8. *What role does love play in decision-making in Interstellar?*
9. *How does trauma shape Tony Stark’s personality and decisions?*
10. *Why is it difficult to distinguish reality from dreams in Inception?*

Overall, the question set provides a comprehensive evaluation framework for testing the ability of different language models to generate accurate, context-aware, and emotionally grounded responses in the movie domain.

6. Models Used

In this project, we used three open-source language models: LLaMA 3, Mistral, and Phi-3. These models were selected to compare their ability to generate meaningful, context-aware responses when combined with a Retrieval-Augmented Generation (RAG) pipeline.

All three models were run locally using Ollama, which allowed us to easily switch between models and evaluate them under the same conditions. Each model received the same input, including the retrieved context and the user’s question, ensuring a fair comparison.

LLaMA 3 is a powerful large language model known for its strong reasoning and ability to generate detailed responses. In our system, it often produced comprehensive answers with good explanations, especially for questions involving character motivations and emotional depth. However, its responses were sometimes longer than necessary (Touvron et al., 2023).

Mistral is a more efficient and lightweight model compared to LLaMA. It provided balanced performance across most questions, generating clear and concise answers while still maintaining reasonable reasoning ability. It performed well in general cases but was slightly less detailed compared to LLaMA in complex reasoning tasks (Jiang et al., 2023).

Phi-3 is a smaller model, but it showed surprisingly strong performance in our experiments. Despite its size, it was able to generate accurate and focused responses when guided by retrieved context. It achieved the best overall performance in terms of accuracy while maintaining good reasoning and clarity (Microsoft, 2024).

By using these three models together, we were able to compare how different architectures and model sizes behave in a RAG-based system. This comparison highlights that even smaller models can perform very well when supported by relevant external knowledge.

7. Results

The performance of the three language models—LLaMA 3, Mistral, and Phi-3—was evaluated using four key metrics: accuracy, reasoning, emotion understanding, and hallucination. The results show that all models performed consistently well in reasoning, achieving a score of 4.0, which indicates strong capability in explaining character motivations and narrative events. Similarly, emotion understanding was also high across models, with both LLaMA 3 and Phi-3 scoring 4.0, while Mistral scored slightly lower at 3.8 (Lewis et al., 2020).

model	accuracy	reasoning	emotion	hallucination
llama	3.0	4.0	4.0	2.0
mistral	2.8	4.0	3.8	2.0
phi	3.2	4.0	4.0	2.0

Figure 4: Results from the 3 Models

In terms of accuracy, Phi-3 achieved the highest score of 3.2, outperforming LLaMA 3, which scored 3.0, and Mistral, which scored 2.8. This indicates that Phi-3 was more precise in generating correct and contextually relevant answers. Despite being a smaller model, Phi-3 demonstrated strong performance when combined with the RAG framework.

All three models showed the same hallucination score of 2.0, which suggests that the use of Retrieval-Augmented Generation effectively reduced the generation of incorrect or unrelated information. This highlights the importance of grounding responses in retrieved data rather than relying solely on model knowledge.

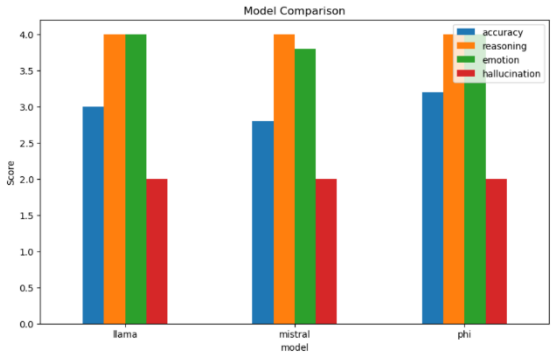

Figure 5: Model Comparison

Overall, the results indicate that while all models benefit from the RAG pipeline, Phi-3 performs the best in terms of accuracy while maintaining strong reasoning and emotional understanding. These findings demonstrate that smaller models can achieve competitive performance when supported by relevant contextual information (Shuster et al., 2021).

8. Analysis

The results show clear differences in how the three models perform within the RAG system. All models achieved strong reasoning and good emotional understanding, which indicates that the retrieved context helped them generate more meaningful answers.

Phi-3 performed the best overall, achieving the highest accuracy while maintaining good reasoning and clarity. Even though it

is a smaller model, it was able to give precise and focused answers. This shows that with RAG, smaller models can perform very effectively.

LLaMA 3 also showed strong reasoning and emotional understanding. It produced more detailed and descriptive answers, but sometimes included extra information, which slightly affected accuracy.

Mistral provided balanced performance, but its accuracy and emotional depth were slightly lower compared to the other models. It performed well for general questions but was less effective in more complex reasoning tasks.

Another important observation is that all models had the same hallucination score. This indicates that the RAG approach successfully reduces incorrect or made-up information by grounding answers in real data.

Overall, while RAG improves all models, the choice of model still matters. Phi-3 achieved the best balance between accuracy, reasoning, and clarity in this project.

9. Discussion

The comparison of the three models highlights that each model has its own strengths and limitations, even when used within the same RAG framework.

Phi-3 demonstrates the strongest overall performance in this project. Its main strength lies in its high accuracy and ability to generate focused, relevant answers. Despite being a smaller model, it effectively utilizes the retrieved context to produce precise responses. However, its limitation is that it sometimes provides less detailed explanations compared to larger models, especially for more complex narrative questions.

LLaMA 3 shows strong reasoning ability and emotional understanding. It is particularly effective at generating detailed and narrative-rich responses, which makes it suitable for explaining complex character motivations and themes. On the downside, its responses can sometimes be verbose, including extra information that may reduce clarity and slightly affect accuracy.

Mistral offers a balanced and efficient performance. It produces clear and concise answers while maintaining reasonable reasoning ability. However, its main weakness is slightly lower accuracy and less depth in emotional understanding compared to Phi-3 and LLaMA 3, particularly for complex psychological or multi-step reasoning questions.

Across all models, one key strength is the reduced hallucination due to the use of RAG. By grounding responses in retrieved data, all models are able to generate more reliable and context-aware answers.

Overall, this analysis shows that while RAG improves the performance of all models, the choice of model still plays an important role. Phi-3 provides the best balance of accuracy and clarity, LLaMA 3 excels in detailed explanations, and Mistral offers efficient but slightly less nuanced responses.

10. Conclusion

In this project, we developed a Retrieval-Augmented Generation (RAG) system for deep movie understanding, focusing on plot

analysis, character motivations, and emotional reasoning. By integrating retrieval with language models, the system was able to generate more accurate and context-aware responses compared to traditional approaches.

We evaluated three open-source models—LLaMA 3, Mistral, and Phi-3—using multiple metrics, including accuracy, reasoning, emotion understanding, and hallucination. The results show that all models benefit significantly from the RAG framework, particularly in reducing hallucination and improving response quality.

Among the models, Phi-3 achieved the best overall performance, demonstrating that even smaller models can produce strong results when supported by relevant contextual information. LLaMA 3 provided detailed and expressive responses, while Mistral offered efficient and balanced performance.

Overall, this project highlights the effectiveness of RAG in enhancing language model capabilities for complex, domain-specific tasks. It also shows that combining retrieval with generation is a powerful approach for improving both accuracy and interpretability in AI systems.

11.Acknowledgement

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